Chest X-ray Abnormalities Detection

Team 2

Group 2

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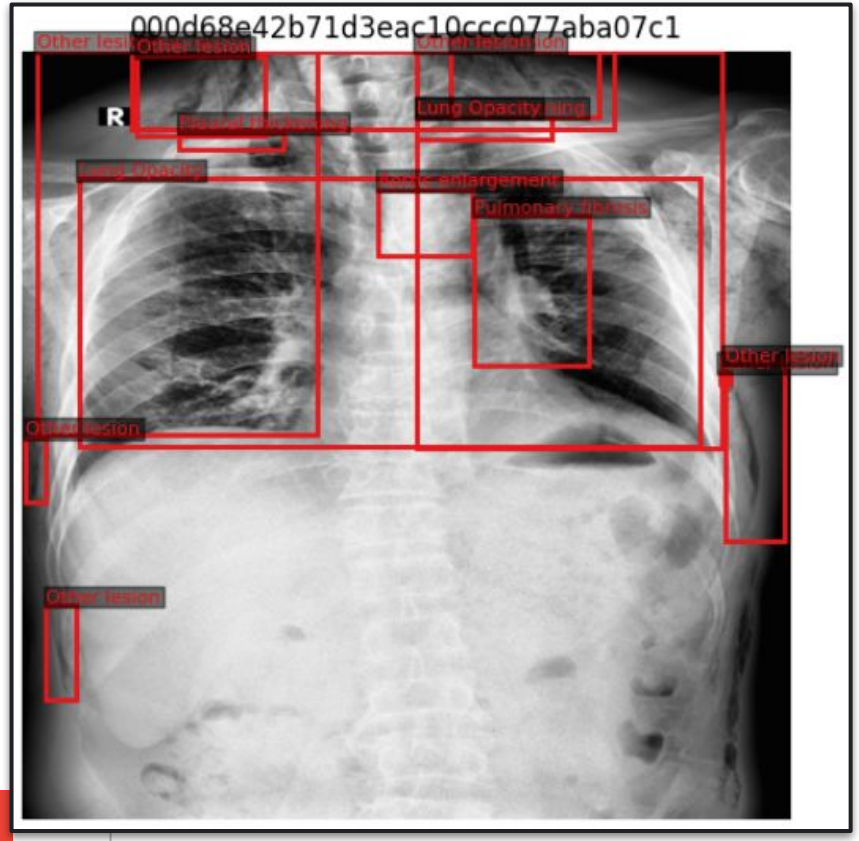
01 | Introduction & Objectives

Introduction

- **Objective:** Develop a high performing detection model for chest X-ray anomaly detection
- **Data:** VinBigData Chest X-ray dataset
- **Application:** Early detection and diagnosis of diseases via visual interpretation

Introduction - Dataset example


000d68e42b71d3eac10ccc077aba07c1
Other lesion
Other lesion
Other lesion
Other lesion
Lung Opacity
Lung Opacity
Lung Opacity

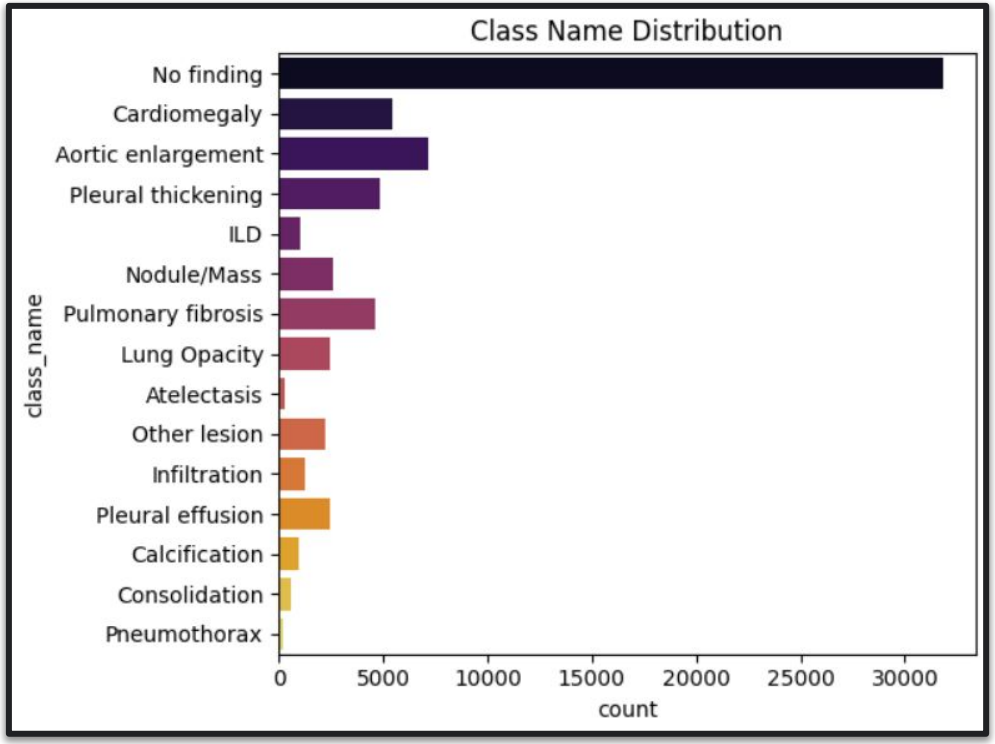
02 | Dataset Overview

Dataset Overview

- Original format: high bit depth (up to 16 bit) DICOM images with pixel array, annotation configurations and patients' metadata
- 18000 total images (15000 annotated training images, 3000 test images)
- 67914 bounding box coordinates with ground-truth anomaly class labels

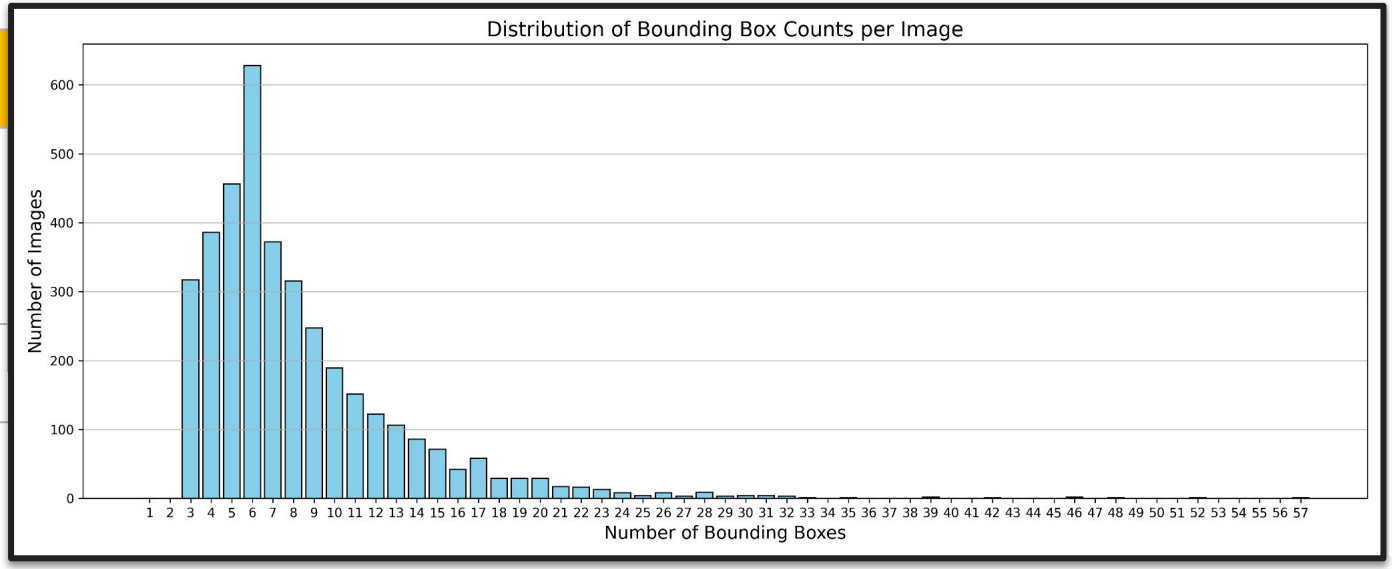
image_id	class_name	class_id	rad_id	x_min	y_min	x_max	y_max
9a5094b2563…	Cardiomegaly	3	R10	691.0	1375.0	1653.0	1831.0

Data Distribution



Imbalance: **10606** images **(67%)** contain no detected anomalies (“No findings”)

Bounding Box Distribution



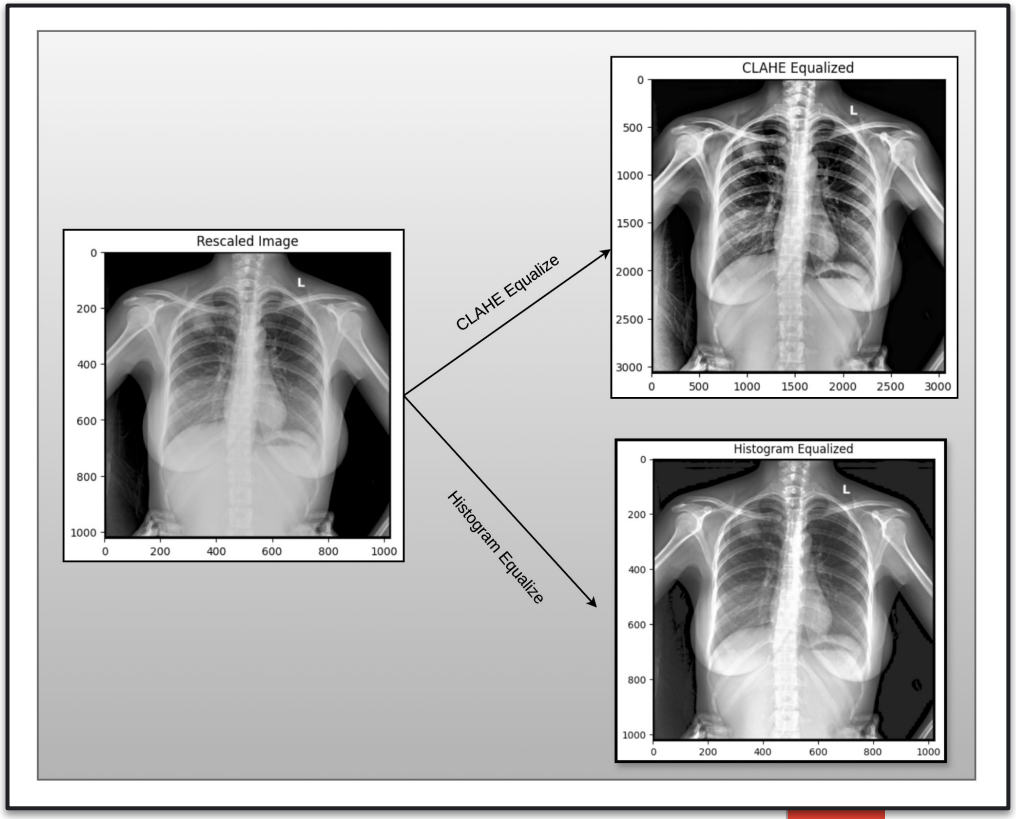
Due to many annotators per image, some abnormalities can be annotated more than once

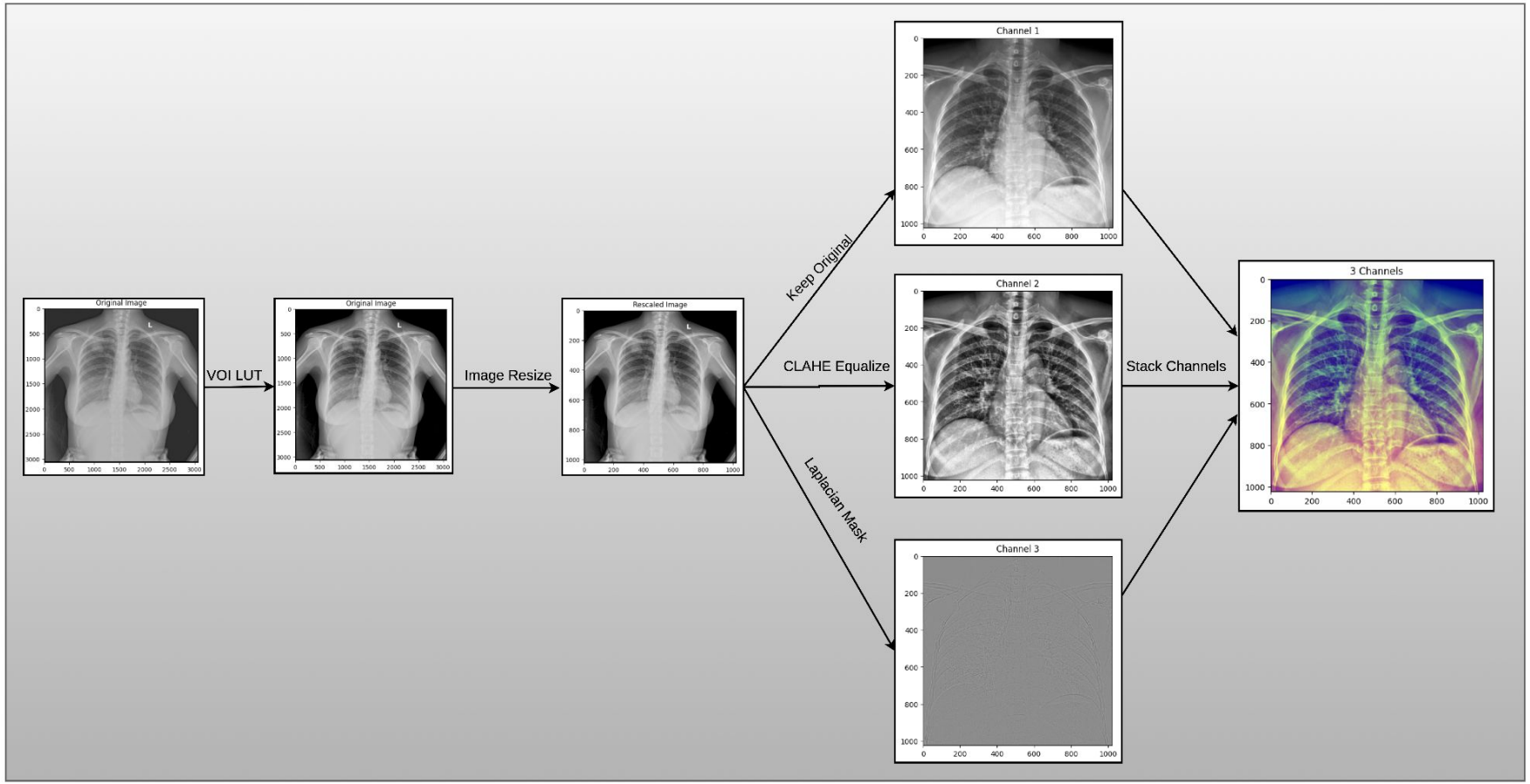
03 | Methodology

Image Preprocessing Pipeline

DICOM to Resized Image	Image Enhancement	Final Dataset
DICOM -> np.array VOI LUT + MONOCHROME Inversion Resize Images to 1024x1024	CLAHE Equalization (1 channel) Base Image + CLAHE + Laplacian mask (3 stacked channels)	The dataset is split into 2 branches (1-channel and 3-channel) for performance comparison

CLAHE vs Histogram Equalization

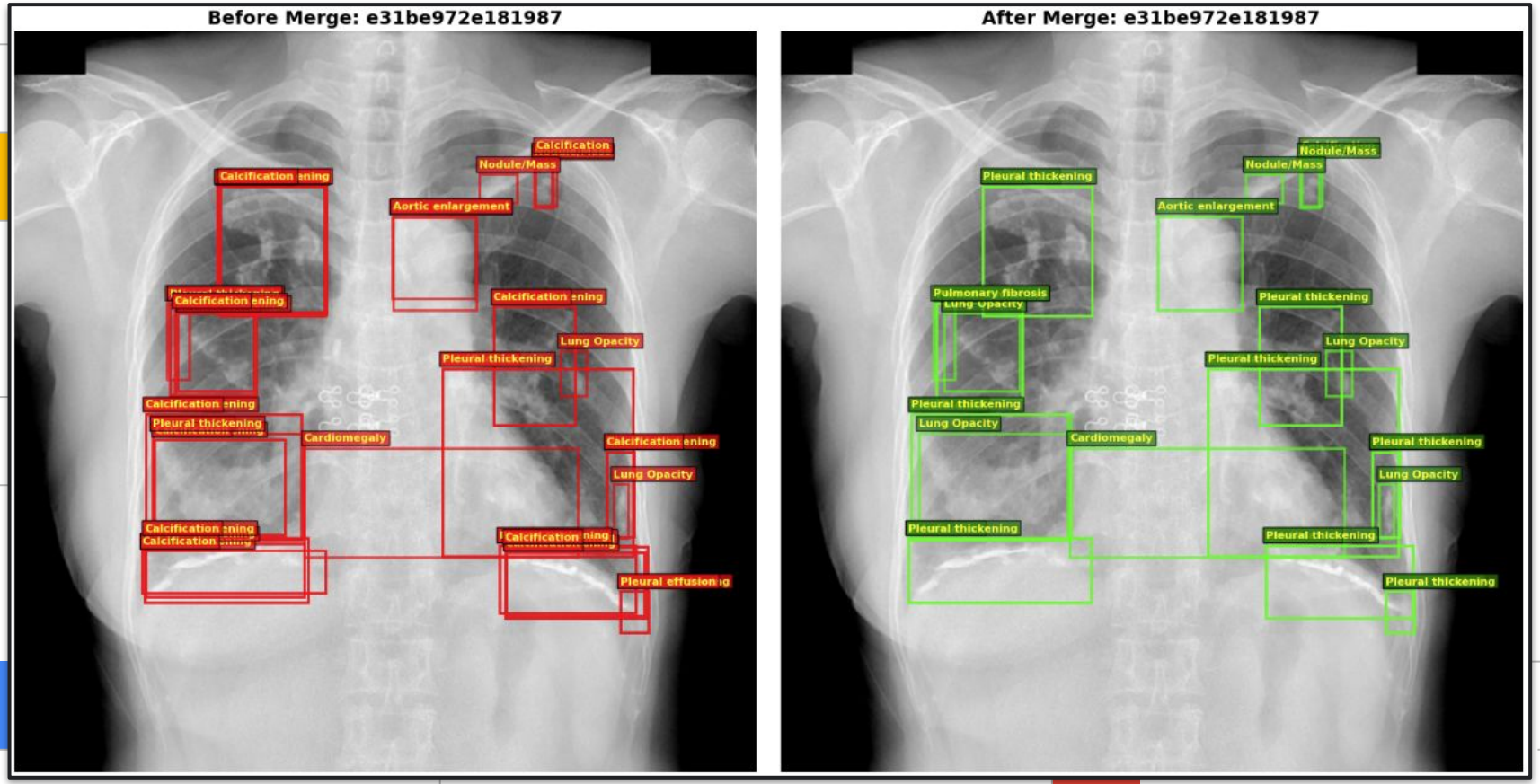



Original Image
VOI LUT
Original Image
Image Resize
Rescaled Image
Keep Original
CLAHE Equalize
Laplacian Mask
Channel 1
Channel 2
Channel 3
Stack Channels
3 Channels

Label Preparation Pipeline

Coordinates Rescaling	Bbox Merging	Negative Handling
Resize bbox coordinates to match 1024x1024 image Calculate center coordinates for YOLO-based models	Merge bboxes with same class + IoU > threshold into a single larger box 67914 (original) -> 32467 (after)	Fill NaN coordinates with 0, which means “No Finding” images will contain all 0 coordinate bbox

Bounding Box Merging



04 | Model Approaches

Applied models & Methodology

- Evaluations split into two branches; 1-channel baseline vs 3-channel

- Models tested:
  - Faster R-CNN (ResNet-50, ConvNeXt V2 backbones),
  - YOLOv8, RT-DETR

- Ensemble strategy: Combined predictions from YOLOv8n and Faster R-CNN trained on 3 channel inputs using **Weighted Boxes Fusion** (WBF)

05 | Results & Discussion

Evaluation metrics

- Standard Object Detection metrics:

  - Precision,

  - Recall,

  - Mean average precision (mAP@0.5, mAP@0.75)

- Free-Response ROC (FROC)

  FROC measures sensitivity relative to false positives per image, providing a more clinically meaningful benchmark for lesion detection

TABLE III
1-Channel Model Evaluation

Model	Precision	Recall	mAP@50	mAP@75	FROC Score
ResNet50-FasterRCNN	0.12	0.26	0.15	0.13	0.29
ConVeXtV2-FasterRCNN	0.14	0.17	0.17	0.14	0.19
YOLOv8n	0.38	0.23	0.22	0.08	0.25
RT-DETR	0.32	0.33	0.27	0.1	0.36

TABLE IV
3-Channel Model Evaluation

Model	Precision	Recall	mAP@50	mAP@75	FROC Score
Faster R-CNN	0.44	0.41	0.3386	0.144	0.4465
YOLOv8n	0.59	0.434	0.4899	0.185	0.5347
WBF Ensemble (Unnormalized)	0.72	0.31	0.4163	0.2136	0.4735
WBF Ensemble (Normalized)	0.73	0.32	0.4202	0.2151	0.4788
WBF Ensemble (Grid Search Skew 0.75)	0.63	0.44	0.4680	0.2278	0.5212

Results & Performance

- 1 channel baseline: YOLOv8n and RT-DETR achieved standard baseline FROC scores (0.25 and 0.36)

- 3 channel advanced evaluation: noticeable improvement over 1-channel
  - Highest FROC: YOLOv8 - 0.5347

- The WBF ensemble model even achieves more robust performance
  - 0.5215 FROC w/ Manual skew 0.75

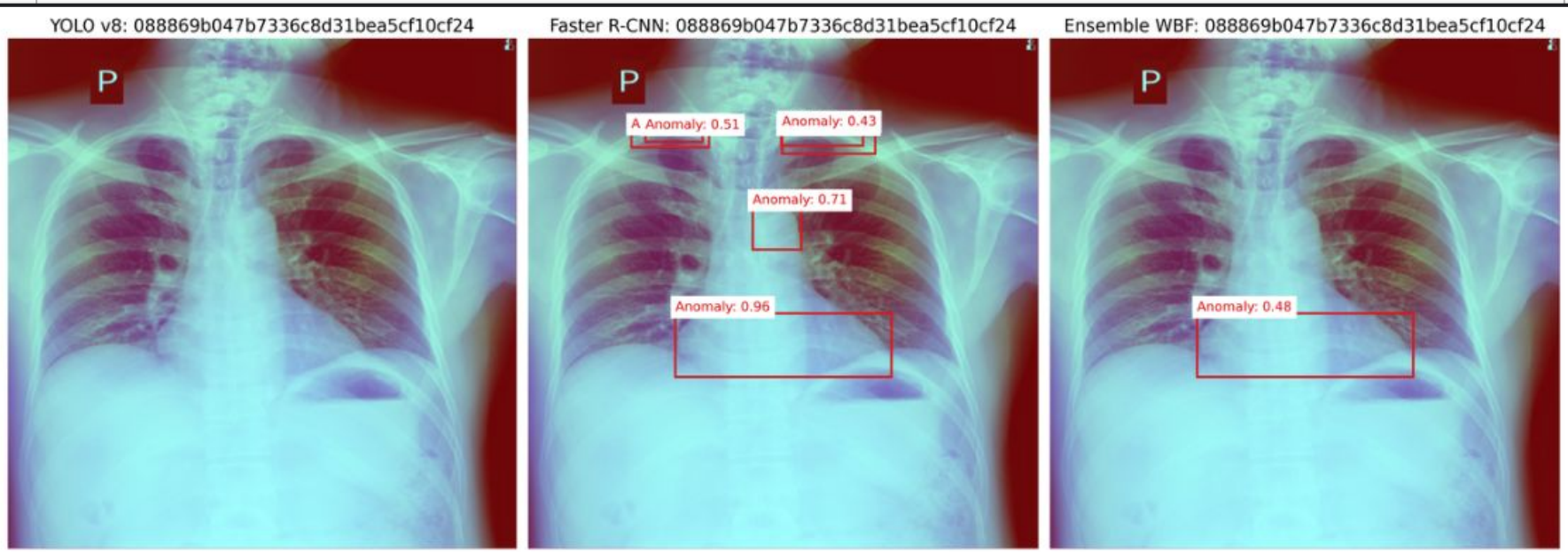
06 | Conclusion

Conclusion

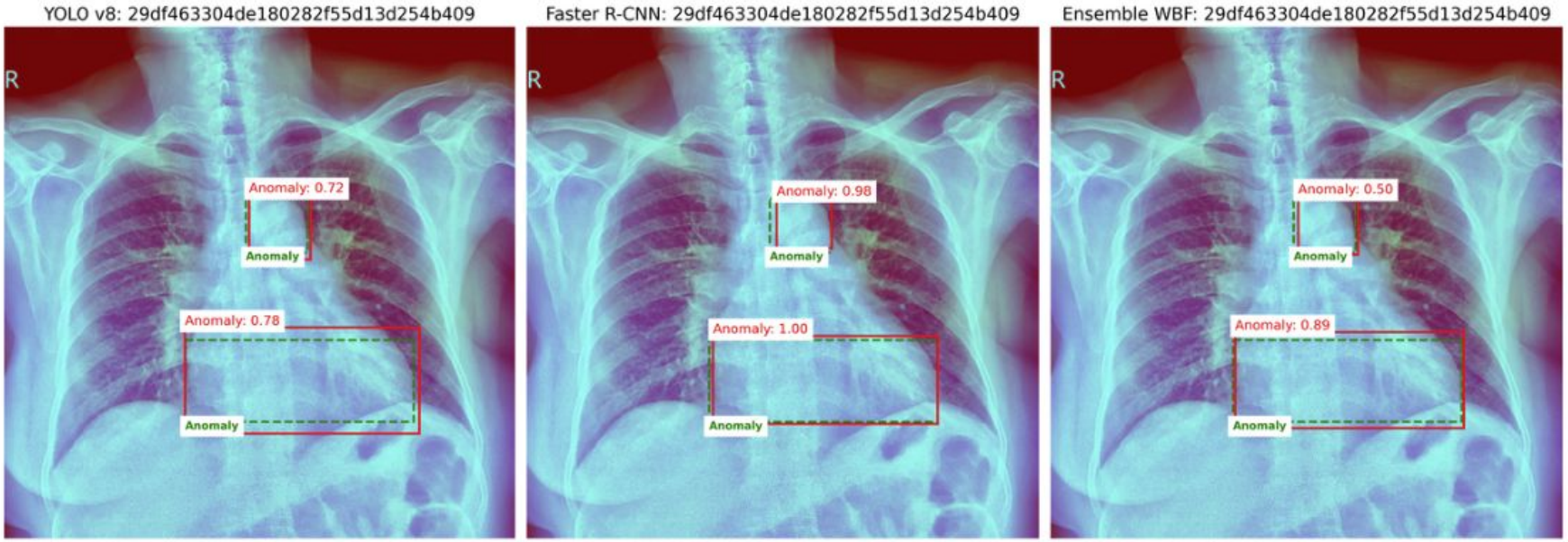
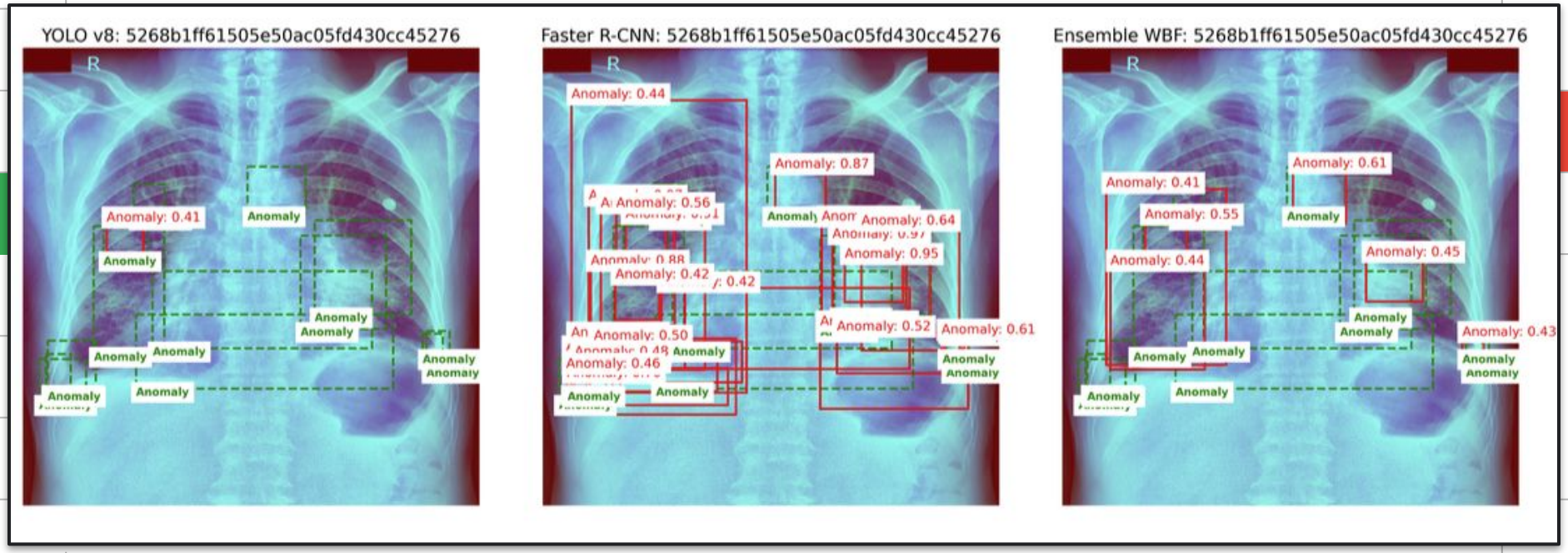
- The 3-channel input pipeline successfully extracts and integrates complementary visual data (raw intensity, contrast-enhanced structural data, and edge definitions).
- Integration of the 3-channel input substantially increases all evaluation metrics compared to basic 1-channel configurations.
- Ensembling YOLO and Faster R-CNN models yields robust generalized performance for chest X-ray anomaly detection.

07 | Appendix

No Finding Ground Truth



Small number of Ground Truth anomalies



Large number of Ground Truth anomalies